

Riemannian Version of Hawking's Theorem

Solution to Exercise 2 (p. 105)

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Abstract

The goal of this exercise is to prove the Riemannian version of Hawking's theorem by adapting the same arguments from the Lorentzian case (Theorem 3.72, [1]) and using their respective Riemannian counterparts.

Theorem 1 (Riemannian Hawking Theorem). *Let (M, g) be a Riemannian manifold satisfying the following conditions:*

- (i) *g is complete,*
- (ii) *there exists some embedded S which separates M as a disjoint union $M = M_- \cup S \cup M_+$, with an infimum $C > 0$ of its expansion towards M_+ , that is, such that its mean curvature vector $\vec{H} = H\vec{n}$, where $\vec{n}$ is the unit normal which points out M_- , satisfies $H \geq C > 0$,*
- (iii) *$\text{Ric}(v, v) \geq 0$ for every v .*

Then $\text{dist}(p, S) \leq 1/C$ for every $p \in M_-$.

Proof. Let us fix an arbitrary point $q \in M^-$. Given that the Riemannian manifold (M, g) is complete by hypothesis (i), we are in a position to apply the Hopf-Rinow theorem. This result acts as the natural counterpart to the Avez-Seifert theorem in the Lorentzian setting, ensuring that the distance from q to the hypersurface S is strictly realized. Let $p \in S$ be the point where this infimum is attained, so that $\text{dist}(q, S) = \text{dist}(q, p)$. We can then trace a length-minimizing geodesic $\gamma : [0, b] \rightarrow M$, parameterized by arc length, connecting p to q . We note that the total length of this curve is precisely $b = \text{dist}(q, S)$.

Since γ is a curve that minimizes the distance between q and the submanifold S , the orthogonality lemma dictates that it must be orthogonal to S at its starting point $\gamma(0) = p$. By the structure of the problem, γ extends into the interior of M^- . Because the statement adopts the usual convention of orienting the normal field $\vec{n}$ towards M^+ , we deduce that the initial velocity vector $\gamma'(0)$ points in the direction diametrically opposite to $\vec{n}$.

Through the study of the second variation of arc length, we know that the global minimizing character of γ forces its index form, evaluated on proper vector fields with fixed endpoints, to be positive semi-definite. By virtue of Theorem 3.24 [1], this is topologically equivalent to the absence of focal points of S along the geodesic restricted to the half-open interval $[0, b)$.

At this stage, the curvature hypotheses come into play. Condition (ii) stipulates that the expansion towards M^+ with respect to the normal frame $\vec{n}$ is bounded from below by $C > 0$, meaning $H \geq C > 0$. Since our geodesic γ is directed towards M^- with initial velocity $\gamma'(0) = -\vec{n}$, its initial expansion is precisely reversed: $H_{\gamma'} = -H \leq -C < 0$. This strict negativity dictates a bounded initial convergence for the geodesics orthogonal to S entering M^- . Pairing this phenomenon with the geometric condition (iii), which yields the spectral non-negativity of the Ricci tensor ($\text{Ric}(v, v) \geq 0$), we assemble all necessary hypotheses for the Riemannian version of the focal point existence theorem (Theorem 3.28, [1]). As a direct consequence, we unavoidably ascertain the emergence of a focal point concerning S for any geodesic delving into M^- at a finite distance $r \leq 1/C$.

To conclude the argument, let us proceed by contradiction and assume there exists some point $q \in M^-$ such that its minimizing geodesic to S satisfies $b = \text{dist}(q, S) > 1/C$. Under such an assumption, the preceding statement dictates the inevitable appearance of a focal point for our curve γ at an instant $r \leq 1/C < b$. Nevertheless, this fact would violate the essential geometric limitation of minimizing geodesics, which must remain free of focal points throughout their entire path $[0, b)$.

This clear contradiction force us to reject our initial assumption that $b > 1/C$. In summary, the distance $\text{dist}(q, S)$ is unconditionally bounded from above by $1/C$ for any conceivable configuration of $q \in M^-$, thereby thoroughly establishing the result. $\square$

References

- [1] Miguel A. Javaloyes Victoria, Miguel Sánchez Caja, *An introduction to Lorentzian Geometry and its applications*, XVI Escola de Geometria Diferencial, (2010)